



Waipapa
Taumata Rau
University
of Auckland

AUCKLAND SITE · AOTEAROA NEW ZEALAND · MON 19 – TUE 20 OCTOBER 2026

How far can a chain of **surrogates** take us in biomolecular interactions?

Biology has a ladder of scales and machine learning now has one alongside it, most of it open-weight. **Two days finding out how far up you have to climb before the answer can be trusted.**

SYSTEMS · PROTEINS · LIGAND BINDING · DRUG DISCOVERY · PHARMACOLOGY

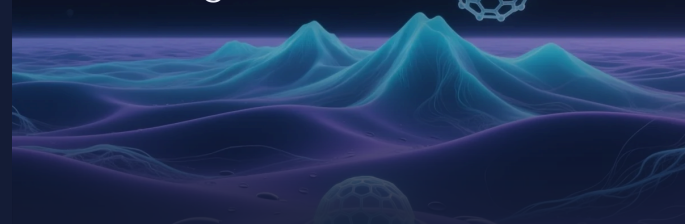
PART OF THE GLOBAL EVENT

 **OSI Hackathon**

The **Open Scientific Intelligence Hackathon**, 4th edition — 120 projects across 16 sites in 2025, and a community paper each year crediting every team. Auckland runs 19–20 October, the global days are 21–22 October, and **their free registration is what puts a project in front of the judges.** luma.com/ku88xh92

THEME 1 MANY MOLECULES, COARSE

Screening at scale

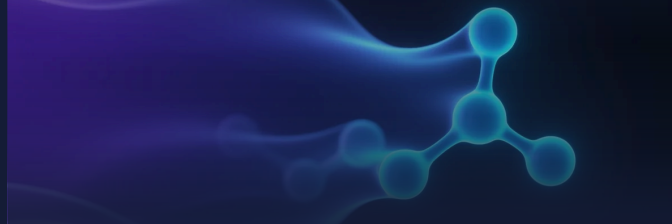


Models that know something about shape, used to filter before you pay for docking.

Boltz-2 · OpenFold3 · IntFold · ChemBERTa-2 · ESM-2

THEME 2 ATOMS, IN MOTION

Molecules in motion

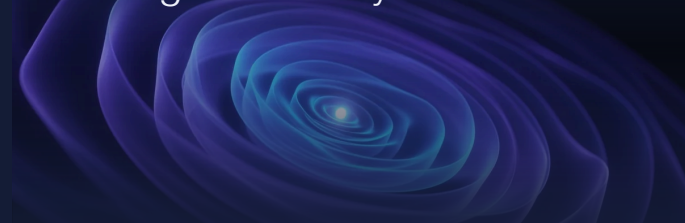


Machine-learned interatomic potentials in MD. Near-quantum forces you can afford to run.

MACE · UMA · eSEN · Orb-v3 · NequIP

THEME 3 ONE EVENT, WHOLE CELL

Binding to whole system



Carrying a molecular signal up to functional and physiological change.

STATE · scGPT · Geneformer · scFoundation

THEME 4 EVERY APPROVED DRUG

Repurposing what we have



Genotype, structure and approved drugs, with deliberately small models.

Evo 2 · Nucleotide Transformer · TxGNN · PrimeKG

Four to start from. Bring your problem and take a space, or propose a theme of your own — the wider event takes agents, datasets, benchmarks, models and software too.

WHO IT'S FOR

Bring your team, your expertise, and a problem you want to look at, or take one of our default targets. Chemists, physicists, structural biologists, statisticians, systems biologists, pharmacologists, clinicians, bioinformaticians, RSEs, and anyone new to it.

You don't need to be able to code. The agent writes the code and you bring the judgement about whether the answer is any good.

No stupid questions · Safe to experiment · A negative is a result

PRACTICAL

When	Mon 19 Oct, 10am – Tue 20 Oct, 4pm
Where	Digital Research Innovation & AI Lab · Level 10, 70 Symonds Street
Cost	Free
Bring	A laptop
Day 1	Get running, report back
Day 2	Measure it, show what you got

Free

All welcome

Cross-disciplinary

Food both days

drai-inn.github.io/osi-hackathon-auckland

Two free steps: RSVP at forms.cloud.microsoft/r/LQCAFcHdZ4, then register with the global event at luma.com/ku88xh92